

D6 The Photoelectric Effect

D6.1 Complete the question in the table:

Wavelength of light /nm	Work function	Max. KE of photoelectrons	Stopping potential /V
650	0.62 eV	(a)	(b)

D6.2 A material will not emit photoelectrons unless it is irradiated by light with a wavelength less than 540 nm. Calculate its work function in electronvolts.

D6.3 A graph of maximum photoelectron kinetic energy (y) in joules against frequency of light (x) in hertz is plotted for a new material. Without knowing more information, answer the following questions:

- What is the significance of the y -intercept?
- What is the significance of the x -intercept?
- What is the value of the gradient?

CHAPTER D. WAVES

Answers

$$\text{D6.1 } K_{\max} = \frac{hc}{\lambda} - \phi = \frac{hc}{650 \times 10^{-9}} - 0.62 \times 1.60 \times 10^{-19} = 2.068 \times 10^{-19} \text{ J}$$
$$V_{\text{stop}} = \frac{K_{\max}}{e} = \frac{2.068 \times 10^{-19}}{1.60 \times 10^{-19}} = 1.2925 \text{ V}$$

$$\text{D6.2 } \phi_{\text{eV}} = \frac{E_{\text{photon, threshold}}}{e} = \frac{hc}{\lambda e} = 2.3021 \text{ eV}$$

D6.3

- a) – work function
- b) threshold frequency
- c) Planck constant